

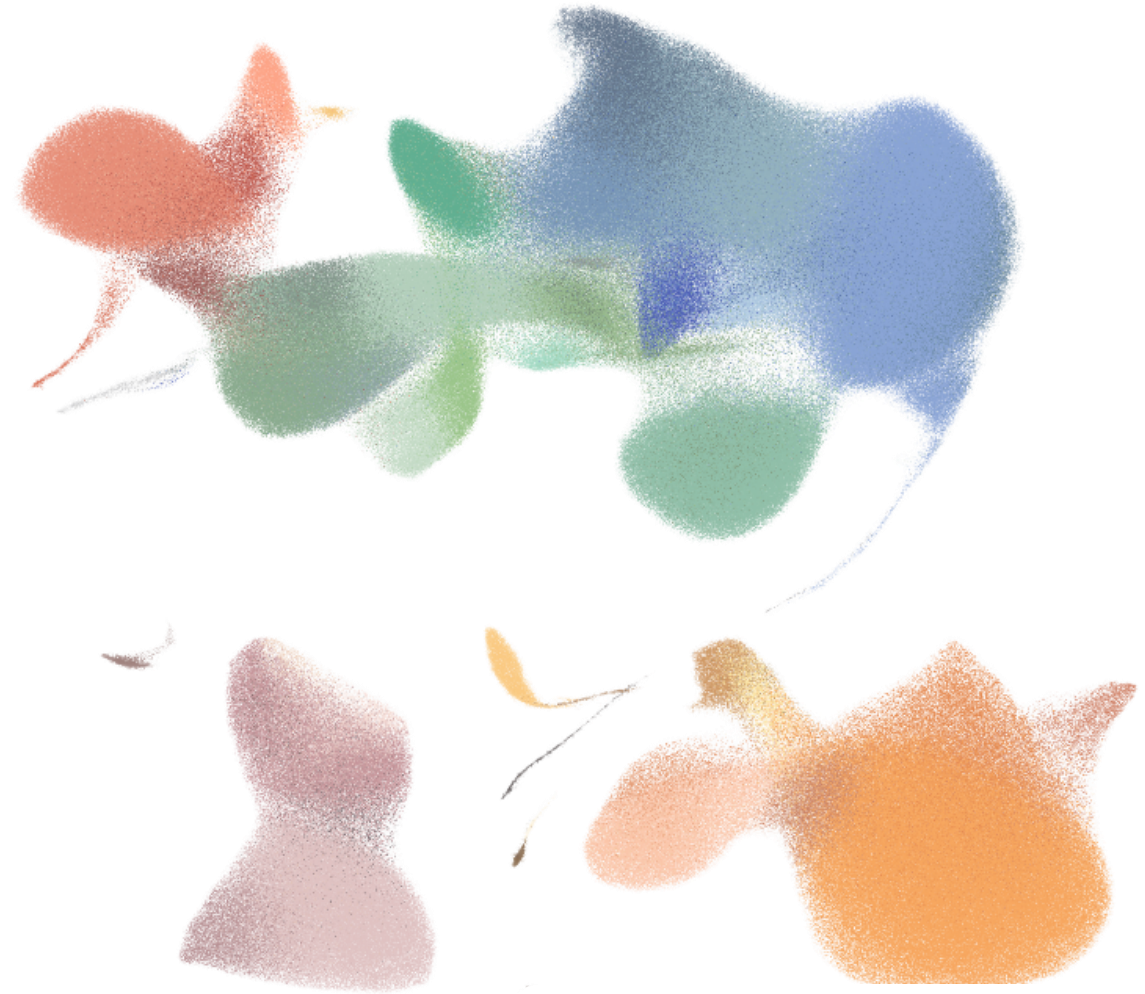
# Single-cell transcriptomics

## COURSE ON SCRNA-SEQ DATA ANALYSIS

What it measures, why it changed biology, and how the data is made

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PART ONE

# What single-cell data is, and why we want it

## THE CORE IDEA

# Bulk RNA-seq gives you an average. Single-cell gives you the distribution.

## Bulk RNA-seq

One value per gene per sample — the mean across every cell in the tube. Like tasting a smoothie: you know berries went in, but not which fruit contributed what.



Bulk: the smoothie — one blended flavour

## Single-cell RNA-seq

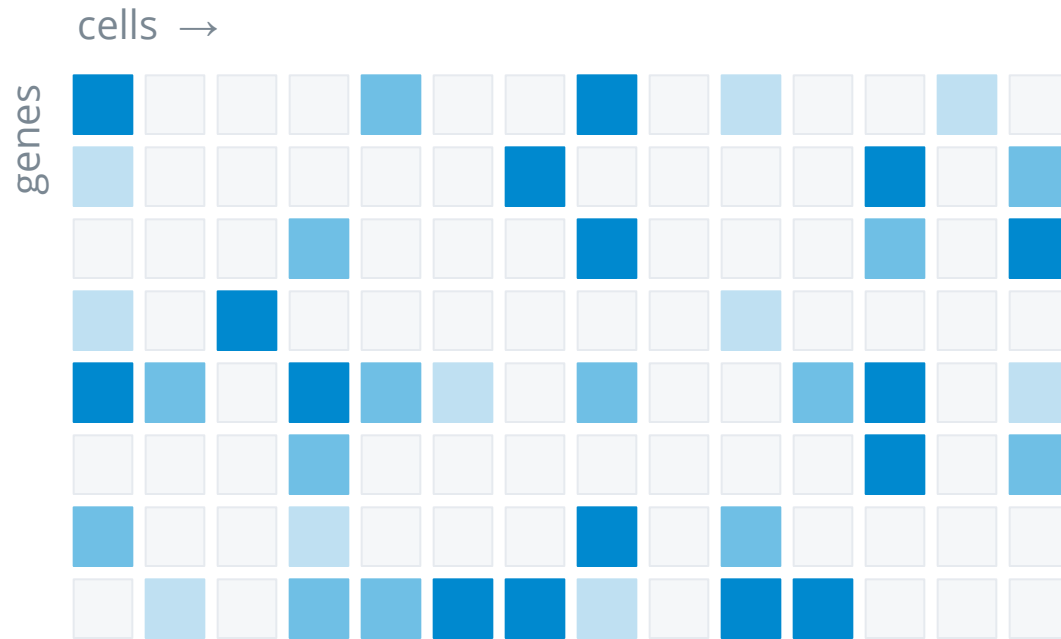
One expression profile per cell — thousands per sample. The fruit stays unblended, so you can still see every berry, count them, and spot the odd one.



Single cell: the berries — each one still separate

## WHAT YOU ACTUALLY GET

The measurement is a matrix of counts — genes × cells — and most of it is zero



integer UMI counts — a zero may mean 'not detected' or 'zero expression'

**~2 000–5 000**

genes detected per cell (typical 3' droplet data)

**80–95 %**

of matrix entries are zero, in typical droplet data

**$10^3$ – $10^6$**

cells per experiment

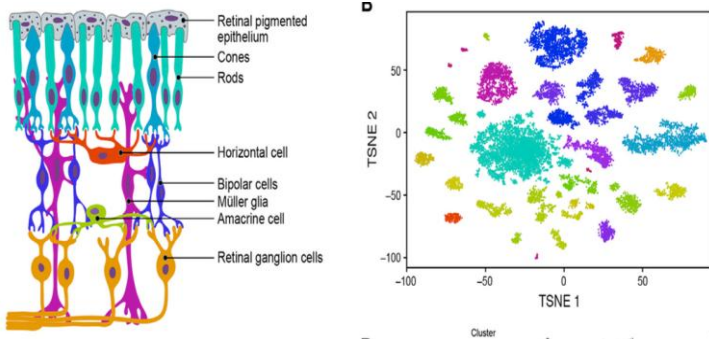
## WHY WE DO IT

# Three questions that only single-cell resolution can answer

01

## Which cell types are in there?

Group cells by their profiles, then name each group. Rare cell types show up too.

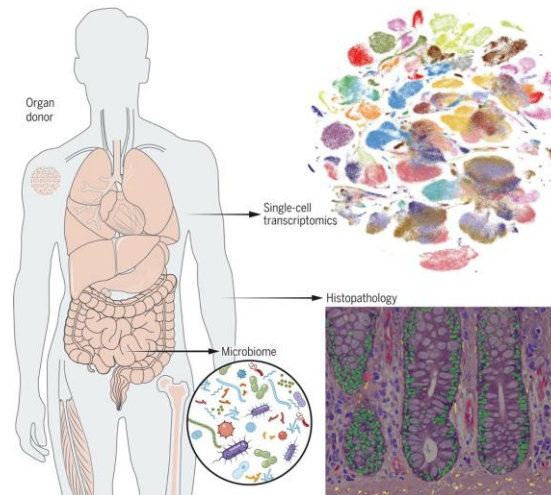


Macosko et al., Cell 2015

02

## How large can the map get?

Dozens of tissues from one donor, all on one map.

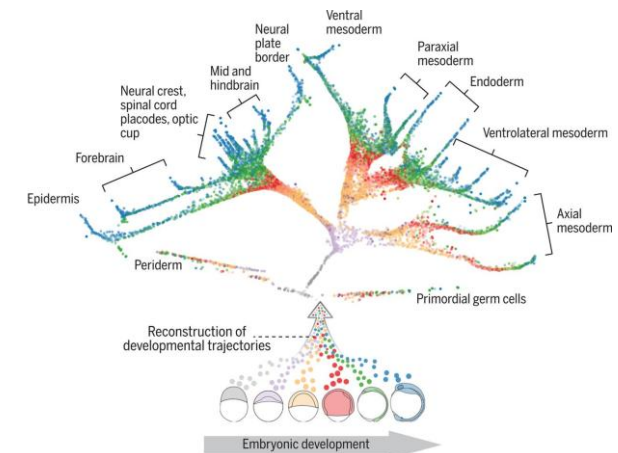


The Tabula Sapiens, Science 2022

03

## How do cells change over time?

Put cells in order along a path and watch one cell type turn into another.



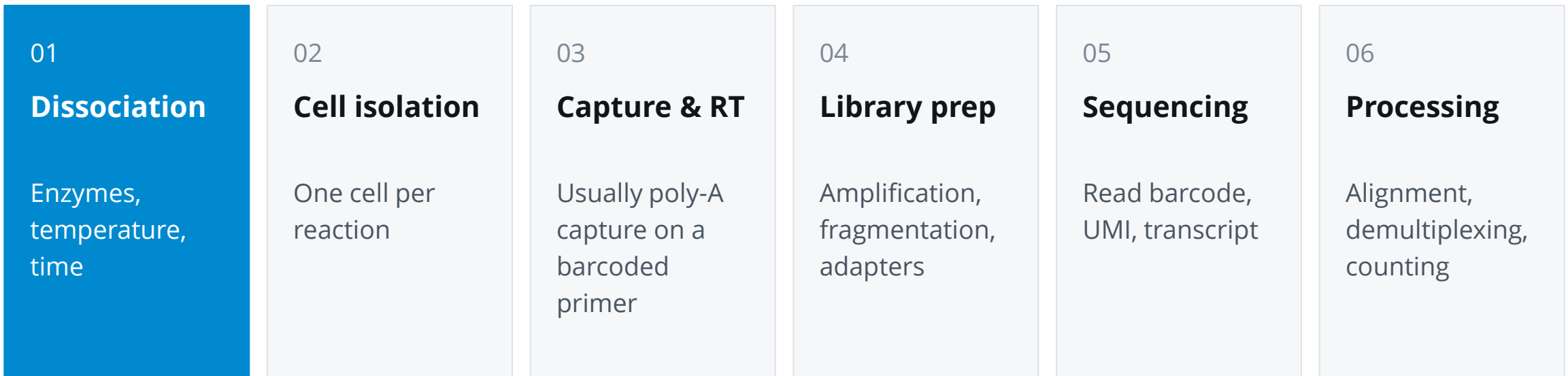
Farrell et al., Science 2018

PART TWO

# How the data is made

## THE WORKFLOW

# Six wet-lab steps stand between a tissue and a count matrix



## Invisible later in the matrix

The matrix keeps no record of a rough dissociation or a stressed sample — it just looks like biology.

## Where this course starts

Step 06 hands you a cells × genes matrix. Everything from QC onwards is what the rest of these lectures cover.

## STEP ONE IN PRACTICE

# Obtaining cells: the step that decides what your data can show

## Yield and quality depend on

### Tissue

From easy (blood, marrow) to effectively impossible (brain, fibrotic or frozen tissue).

### Lab

Which instrument you have — and how far it is from the bench. Minutes matter.

### Handling

Gentle, cold, fast. Stress responses are transcribed while you work.

## A usable suspension is

### Clean

no clumps, no free-floating debris

### Healthy

high viability, low ambient RNA

### Intact

membranes whole, no decomposition

FACS or MACS enrichment before loading is fair game — and often the only way to see a rare population at reasonable cost.



PART THREE

# Technologies and methods

## FOUR FAMILIES

# Every protocol solves the same problem: get one barcode into one cell

### Plate-based

#### Smart-seq2/3

One cell per well, sorted or picked. Full-length coverage, hundreds of cells.

**$10^2$ – $10^3$  cells**

### Droplet-based

#### 10x Chromium, Drop-seq

Cell and barcoded bead co-encapsulated in an oil droplet. 3' (or 5') counting.

**$10^3$ – $10^5$  cells**

### Combinatorial indexing

#### SPLiT-seq, sci-RNA-seq

No physical isolation — cells get a unique combination of barcodes over split-pool rounds.

**$10^4$ – $10^6$  cells**

### Microwell / nanowell

#### BD Rhapsody, Seq-Well

Cells settle into an array of wells with barcoded beads. No microfluidics.

**$10^3$ – $10^4$  cells**

Throughput and cost per cell have driven the field: droplet methods now dominate because they buy cell numbers, which is what most biological questions need.

## FAMILY ONE

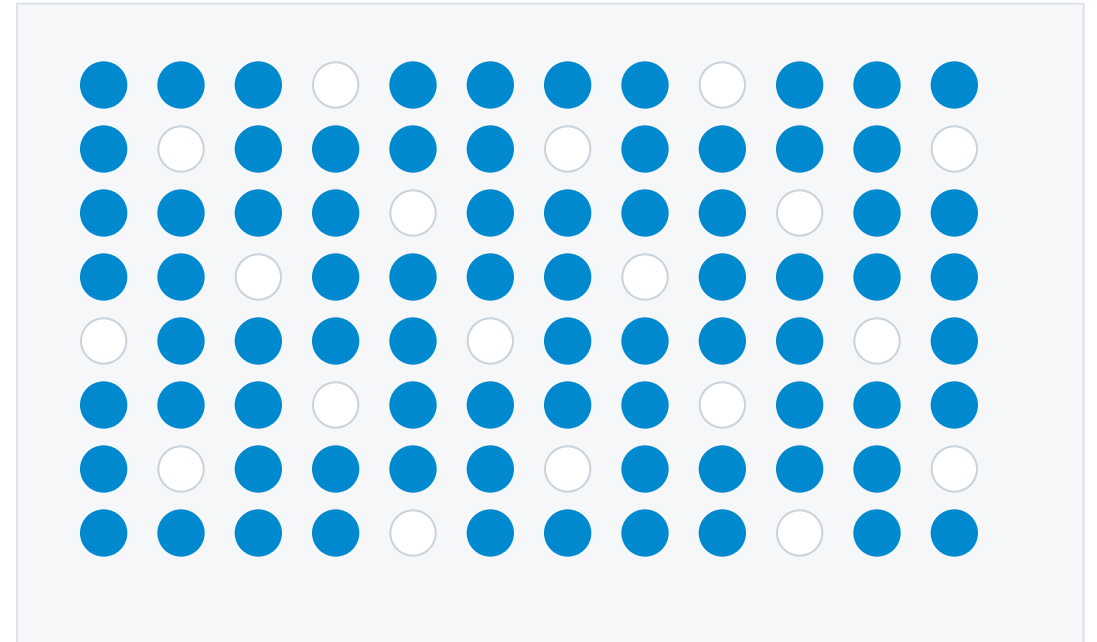
# Plate-based: one cell per well, sequenced deeply

## How it works

Single cells are sorted or picked into individual wells, lysed, and reverse-transcribed with a template-switching oligo. Each well is a separate library, so the whole transcript is covered — not just its 3' end.

## What that gets you

Isoforms, splice junctions and allele-specific expression. Highest sensitivity per cell — thousands more genes detected than in droplet data. Smart-seq3 adds full-length plus UMIs.



96 or 384 wells per plate — one cell each

Trade-off: hundreds of cells, days of hands-on work, and a per-cell cost is a lot higher than droplet-based.

## FAMILY TWO

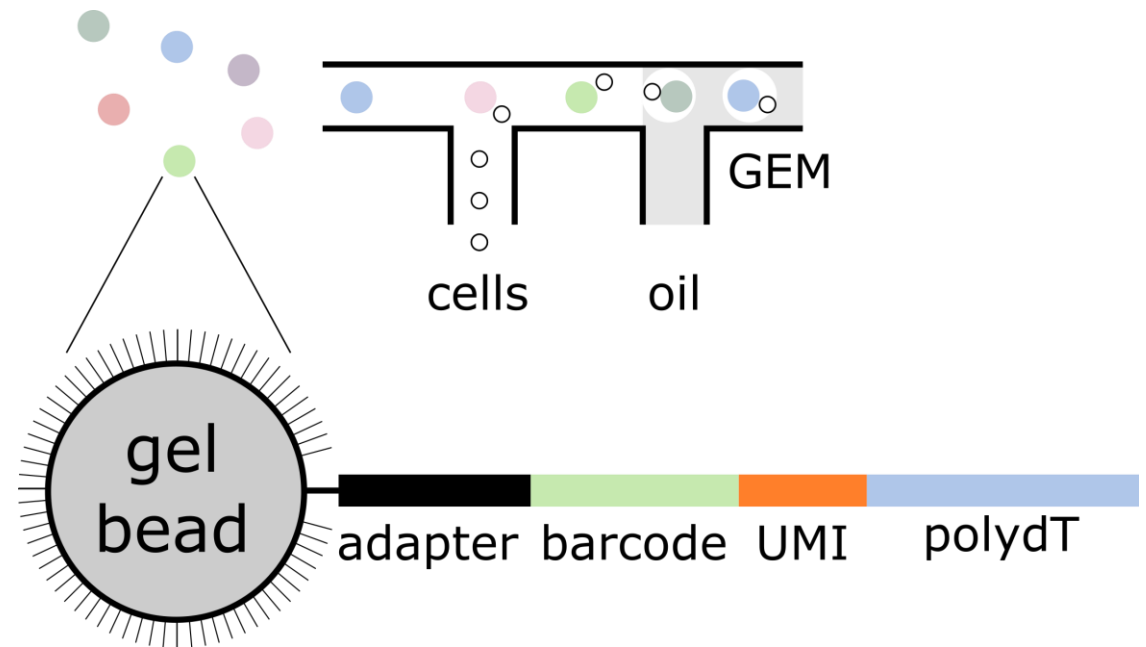
# Droplet-based: one cell and one barcoded bead per droplet

## How it works

Cells, barcoded gel beads and oil meet at a microfluidic junction. Each droplet holds one cell and one bead carrying millions of primers with the same cell barcode plus a unique UMI per molecule. Lysis and reverse transcription happen inside the droplet; then everything is pooled.

## What to keep in mind

Loading is Poissonian: load more cells and you get more doublets. Only the 3' end is sequenced, so isoforms are out of reach, and ~2–5 k genes per cell is typical.



Schematic: 10x Genomics Chromium GEM formation

## FAMILY THREE

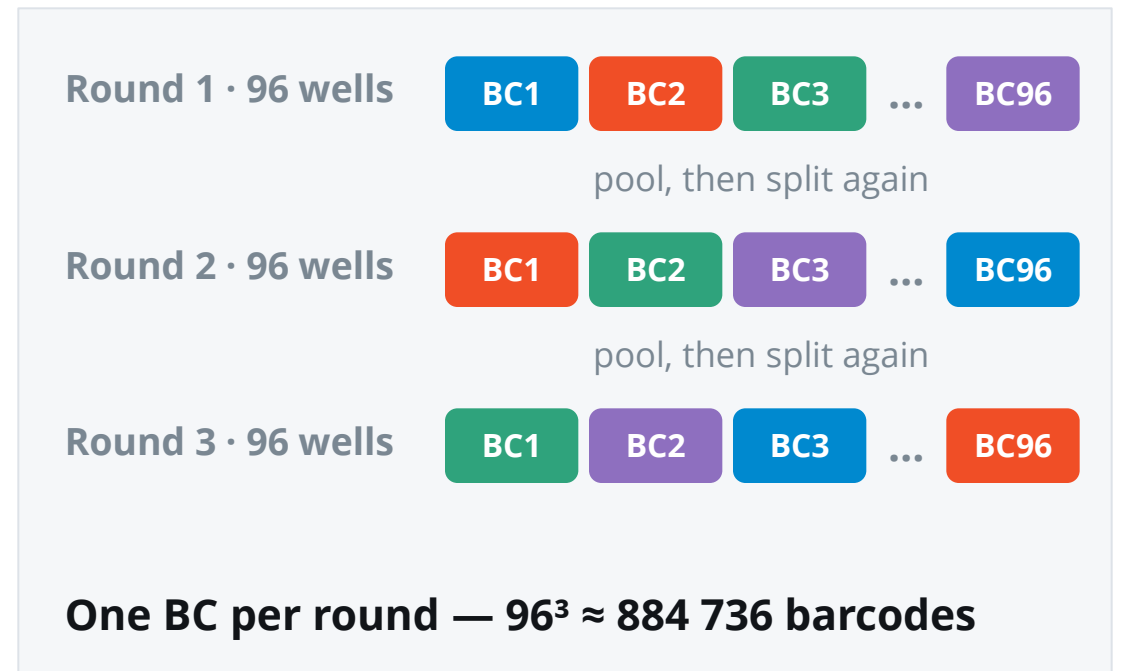
# Combinatorial indexing: no isolation at all, just a barcode address

## How it works

Cells are split across a plate and each well adds a barcode. The cells are pooled, split again, and barcoded again. After four rounds, the chance that two cells travelled the same route is negligible — so the barcode combination identifies the cell.

## Why it matters

No instrument, no microfluidics. Scales to millions of nuclei — this is how whole-embryo and whole-organism atlases are built.



SPLiT-seq: Rosenberg et al., Science 2018

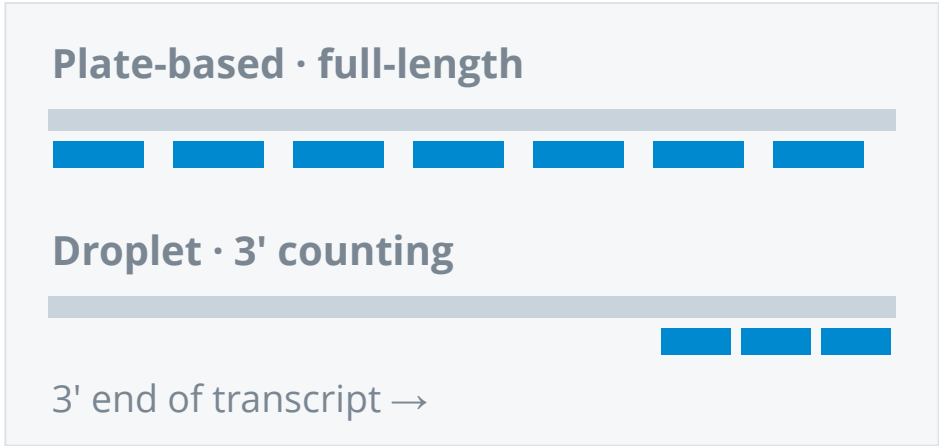
CHOOSING A METHOD

# The trade-off is always the same: more cells, or more per cell

|                     | Plate-based                      | Droplet-based                    |
|---------------------|----------------------------------|----------------------------------|
| Cells per run       | 10 <sup>2</sup> –10 <sup>3</sup> | 10 <sup>3</sup> –10 <sup>5</sup> |
| Genes per cell      | 8 000–12 000                     | 2 000–5 000                      |
| Transcript coverage | full length                      | 3' (or 5') end only              |
| Isoforms, SNPs      | yes                              | no, SNP only in sequenced end    |
| Cost per cell       | high                             | low                              |

## Read depth per cell

Full-length protocols cover the whole transcript; droplet protocols pile reads at one end. Same gene, very different information.



blue bars = sequencing reads

## EXTENSIONS

# The same barcode trick carries more than mRNA

 **CITE-seq**  
Surface protein + RNA

Antibodies tagged with sequencable oligos targeted to surface proteins are counted alongside transcripts, so phenotype and expression come from the same cell.

 **Cell hashing**  
Many samples, one run

Each sample gets its own oligo-tagged antibody against common surface proteins, samples are pooled, then demultiplexed in silico — cheaper, and batch effects collapse.

 **snRNA-seq**  
Nuclei instead of cells

For tissue that will not dissociate — brain, muscle, frozen archives. Only nuclear RNA is captured; fewer transcripts per cell and a large fraction of intronic reads.

## BEFORE YOU START

# Design decisions you cannot fix in the analysis

### Replicates

Cells are not replicates. Statistical power for a condition comparison comes from donors or animals — budget for several per group.

### Batches

Process samples together or pool them with hashing. If you need multiple runs, put every condition in every run.

### Cells vs depth

Rare population? Buy more cells. Isoforms or subtle regulation? Buy more depth.

### Tissue handling

Decide dissociation, fixation and freezing before the first sample arrives — and keep it identical.

## What to take away

Single-cell data is a sparse count matrix produced by a long wet-lab chain. The method you pick sets the ceiling on what the analysis can find.



# Thank you

COURSE ON SCRNA-SEQ DATA ANALYSIS

